

Resilient Multimodal Industrial Surface Defect Detection With Uncertain Sensors Availability

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Abstract—Multimodal industrial surface defect detection (MISDD) aims to identify and locate defect in industrial products by fusing RGB and 3-D modalities. This article focuses on modality-missing problems caused by uncertain sensors availability in MISDD. In this context, the fusion of multiple modalities encounters several troubles, including learning mode transformation and information vacancy. To this end, we first propose cross-modal prompt learning, which includes: 1) the cross-modal consistency prompt serves the establishment of information consistency of dual visual modalities; 2) the modality-specific prompt is inserted to adapt different input patterns; and 3) the missing-aware prompt is attached to compensate for the information vacancy caused by dynamic modalities-missing. In addition, we propose symmetric contrastive learning, which utilizes text modality as a bridge for fusion of dual vision modalities. Specifically, a paired antithetical text prompt is designed to generate binary text semantics, and triple-modal contrastive pretraining is offered to accomplish multimodal learning. Experiment results show that our proposed method achieves 73.83% I-AUROC and 93.05% P-AUROC with a total missing rate 0.7 for RGB and 3-D modalities (exceeding state-of-the-art methods 3.84% and 5.58% , respectively), and outperforms existing approaches to varying degrees under different missing types and rates.

Index Terms—Industrial automation, multimodal learning, sensor availability, surface defect detection.

I. INTRODUCTION

INDUSTRIAL surface defect detection (ISDD) [1], [2], [3] assures quality control in manufacturing and production

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The source code will be available at <https://github.com/SvyJ/MISDD-MM>.

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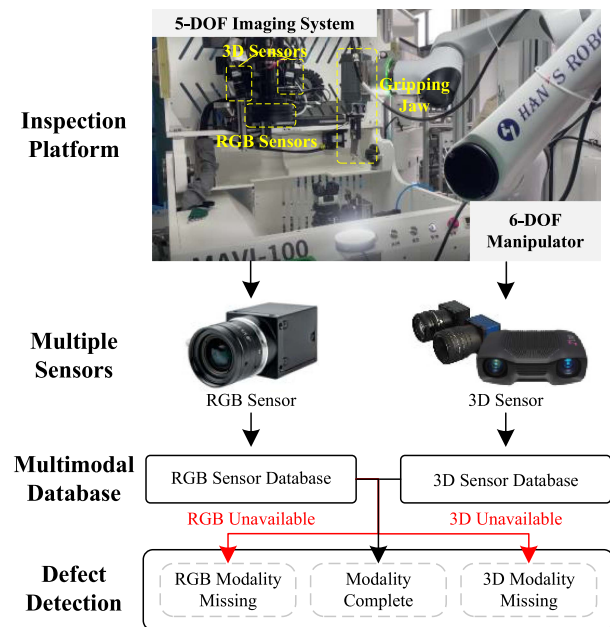


Fig. 1. Modalities missing scenarios caused by the uncertain availability of multiple sensors.

systems by identifying data features that deviate from normal patterns. Due to the scarcity of negative samples, ISDD typically manifests as an unsupervised paradigm. In the past few years, researchers have devoted much effort and greatly advanced the progress of unsupervised ISDD [2], [4], [5]. However, early research was mainly limited to the RGB modality, which was not sufficient for applying to capture defective targets with geometric features [6], [7]. This limitation has driven the emergence of multimodal industrial surface defect detection (MISDD), it supplements the necessary geometric priors by introducing 3-D modalities such as point clouds [8], [9], depth [10], [11], or surface normals [12].

The transition from single-modal [1], [2], [4], [13], [14] to multimodal [9], [15], [16] enables more accurate and comprehensive surface defect detection. But unexceptionally, multimodal learning including MISDD faces a common key issue, that is, the availability of various sensors. As shown in Fig. 1, the automatic surface defect inspection platform [17] we have developed includes automatic imaging and acquisition of RGB and 3-D modalities. Specifically, during the training or inference process, one or some modalities may be missing due

to sensor failures, environmental limitations, or cost control. In the practical dynamic industrial environment, the problem of modalities missing can be exacerbated due to unpredictable changes in sensors availability. To this end, Sui et al. [18] and Miao et al. [19] proposed the modality-incomplete industrial surface defect detection (MIISDD) task, which focuses on static data incompleteness (inherent modality missing of training and testing data), emphasizing pseudo modality imputation of modal features and representation learning. In contrast, we propose MISDD with Missing Modalities (MISDD-MM), which considers dynamic and sample-specific modality absence during both training and inference, as can be seen from Fig. 2. MISDD-MM is defined as a practical multimodal learning scenario where the availability of RGB and 3-D inputs may fluctuate unpredictably due to sensor malfunctions, environmental conditions, or adaptive acquisition strategies. This setting requires the model to perform robust defect localization and adaptive modality fusion under incomplete and uncertain input conditions, as illustrated in Fig. 1. Therefore, the multimodal learning of MISDD-MM would be subject to two main troubles.

First, dynamic multimodal input requires the model to be able to cope with different missing types of data, which demands the ability of multimodal learning mode transformation and processing information vacancy, this is what existing multimodal models undoubtedly lack. Simply, the most direct approach is to replace or impute the missing modality, which usually leads to information distortion or modal illusion. On the contrary, it is more desirable to enable the model to adapt to the problem of MM. Recently, prompt learning has been proven effective in vision-language learning [20], [21] with incomplete modalities, it leverages a few learnable parameters and optimizes them in the middle layer of the transformer through fine-tuning. But differently, the MISDD-MM task relies on dual visual modalities, requiring the model to have the ability to deal with dynamic visual inputs. To this end, we propose the cross-modal prompt learning (CPL), which includes: 1) cross-modal consistency prompt (CCP) for building multimodal visual information consistency for surface defect localization; 2) modality-specific prompt (MSP) for adapting learn mode transformation under dynamic input patterns, to safeguard the feature extraction of a single modality; 3) missing-aware prompt (MAP) for compensating information vacancy caused by dynamic modalities-missing, thus assisting cross-modal interaction.

Second, dual visual modalities cannot be directly fused if one of them is missing. Alternatively, one feasible method is to map them to a shared latent space through contrastive learning. Therefore, we propose to utilize text modality to serve as a bridge for the fusion of RGB and 3-D modalities. In addition, text prompts can produce both normal and abnormal patterns simultaneously [22], [23], [24], which would greatly benefit MISDD-MM. However, the most commonly adopted visual language model (VLM), such as CLIP [25] and ViLT [26], lacks the strength to directly process triple modalities. To cater to this special requirement, we propose symmetric contrastive learning (SCL), which first introduces antithetical text prompts to generate binary text semantics from normal-only samples.

Next, the triple-modal contrastive pre-training is considered as two symmetric visual-language learning with binary prompts.

In summary, by utilizing frozen feature extractors and in cooperation with learnable prompts in CPL and SCL, the solution to the MISDD-MM task is provided. Our contributions are summarized as follows.

- 1) We introduce a new challenging task that caters more to real-world data acquisition, i.e., MISDD with MM caused by uncertain sensors availability.
- 2) We design a resilient MISDD-MM framework that integrates CPL, enabling multimodal transformer to adapt learning mode transformation and information vacancy caused by modalities-missing.
- 3) We propose the SCL paradigm for triple-modal contrastive pretraining, enhancing multimodal fusion with guidance of antithetical text prompt.
- 4) Our proposed method achieves state-of-the-art MISDD-MM performance under different settings of missing types and missing rates.

II. RELATED WORK

A. MISDD

Currently, the mainstream MISDD methods leverage two modalities as inputs, i.e., RGB images and point cloud data (or depth maps). According to technical details, state-of-the-art MISDD methods can be divided into three types: 1) *Memory bank-based* methods [1], [8], [9], [18] detect defects by storing normal sample features and calculating the similarity between subsequent samples and bank. 2) *Teacher-student architecture-based* methods [11], [27] force the pretrained complex teacher model to guide lightweight student models to learn feature representations of normal samples through knowledge distillation. 3) *Reconstruction-based* methods [10], [16], [28] typically implement defect detection by reconstructing specific images, such as from synthetic defective samples to normal samples, from 2-D to 3-D images, etc. The abovementioned methods are based on the premise of modality-complete, which is not applicable to real-world data acquisition scenarios. Therefore, this article introduces a more practical task termed MISDD with MM.

B. Visual Prompt Learning

Prompt learning has efficiently transferred from natural language processing [29] to computer vision [30] and multimodal learning [31]. Early visual prompt learning utilized the cross-modal alignment ability of pretrained VLMs such as CLIP [25] and ViLT [26]) by optimizing text prompts [32] or jointly optimizing visual and text prompts [20]. Static prompt insertion [33] and dynamic prompt generation [34] represent two different types of prompt learning, the former directly changing image feature representation, while the latter adaptively adjusting prompts based on input content, enhancing the adaptability to diverse scenarios. Recently, cross-modal prompt [35], [36] framework have also been constructed. Intuitively, the application of prompt learning in multimodal tasks is mostly

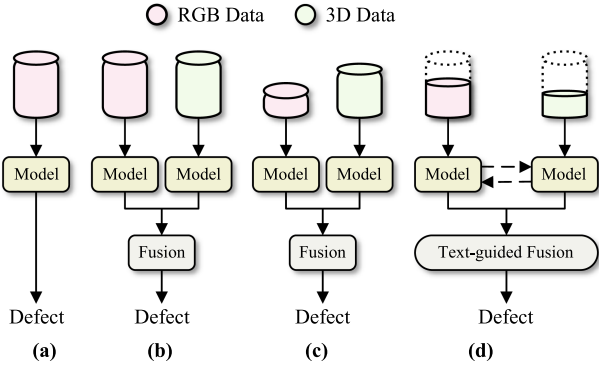


Fig. 2. Task differentiation diagram. Our proposed MISDD-MM task demands multimodal learning under dynamic missing of RGB and 3-D modalities, which differs from static modality-incomplete in MIISDD. (a) ISDD. (b) MISDD. (c) MIISDD. (d) MISDD-MM.

concentrated in VLMs. However, in fact, its scalability in more modalities has also been widely verified, such as video [37], audio [38], 3-D image modalities (such as point clouds [39] and depth maps [40]), etc. Similarly, this article aims to establish a cross-modal prompt mechanism between RGB and 3-D modalities to support multimodal learning for industrial surface defect detection.

C. Multimodal Learning With Missing Modality

Implementing multimodal learning in situations where one or some modalities are missing has more practical application prospects, but it is also highly challenging. Straightforward solutions such as modality imputation [41] and estimation [42] are prone to introduce noise, while shared representation learning [15], [43] directly processes incomplete data through joint embedding. Recently, prompt learning has emerged as a powerful paradigm for handling MM [21], [44], offering low-cost adaptation [34] and cross-modal knowledge transfer advantages [19]. Hence, this article further explores the prospects of leveraging prompt learning in the task of MISDD with MM.

III. APPROACH

A. Overall Framework

Fig. 3 presents the overall flowchart of our proposed method, which consists of three phases, i.e., MM configuration, CPL, and SCL. Specifically, three input patterns are generated based on the MM configuration, where the modality-missing data is initialized by zero padding at the input-level or feature-level, termed as dummy input. Note that as the 3-D modality is either a point cloud or depth map, so there is an optional projection operation to obtain depth map from the point cloud data. Subsequently, three paired cross-modal prompts are attached to the transformer layer and are optionally inherited with the increase in depth. On the one hand, they serve as MAPs for different inputs, and on the other hand, they are utilized for adapting to feature extraction of a single modality. In addition, the galleries placed on both sides are used to store the RGB and 3-D features of normal samples during the training process. Next, normal

text prompts and synthetic abnormal text prompts are fed into a text tokenizer to generate semantic duality, and then perform interaction with visual features. This process is implemented through our proposed SCL, which guides to narrowing the distance between normal samples and normal text prompts, while increasing that between abnormal text prompts. Similarly, a text feature gallery is established to store semantic duality for the inference process.

B. MM Configuration

In this article, three different modalities, i.e., RGB, 3-D, and text, are leveraged, but only RGB and 3-D modalities collected from sensors are simulated to be missing due to uncertain availability of sensors. Accordingly, there are three different missing types, i.e., RGB missing, 3-D missing, and both missing, resulting in three distinct data patterns, i.e., RGB-only, 3-D-only, and modality-complete. Both modalities missing may appear in the dataset simultaneously, but not for each sample. To simulate such cases, the modality-missing data is replaced by dummy input, which can be initialized through the following process.

Modality missing may occur at the input level, as well as at the feature level, which we also refer to as pseudomissing. Feature-level missing is specially designed to accommodate memory bank-based methods such as M3DM [8] and shape-guided [9]. For the sake of unification and comparison, we have set the type and pattern of modality missing to be only dependent on the sample and not on the position of missing, and the data pattern is determined with inputs. Specifically, different data patterns are labeled with an indicating matrix $\mathcal{M} \in \mathbb{R}^{C \times H \times W}$, where C , H , and W denotes the channels, height, and width of input or extracted feature maps. The definition of $\mathcal{M}$ is according to the following rules.

$$\langle \mathcal{M}_i^{rgb}, \mathcal{M}_i^{3D} \rangle = \begin{cases} \langle \mathbf{0}, \mathbf{1} \rangle, & \text{if } \mathcal{D}_i^{rgb} \text{ is missing} \\ \langle \mathbf{1}, \mathbf{0} \rangle, & \text{if } \mathcal{D}_i^{3D} \text{ is missing} \\ \langle \mathbf{1}, \mathbf{1} \rangle, & \text{otherwise.} \end{cases} \quad (1)$$

1) *Input-Level Missing*: Modality missing at the input level is the main simulation method in this article, as it is more in line with actual data acquisition scenarios. Concretely, given a paired RGB image $\mathcal{D}_i^{rgb}$ and 3-D data $\mathcal{D}_i^{3D}$, according to the data missing rules in (1), the model input $\tilde{\mathcal{X}}_i$ generated by the input-level missing scheme can be formulated as

$$\tilde{\mathcal{X}}_i = \langle \mathcal{D}_i^{rgb} \otimes \mathcal{M}_i^{rgb}, \mathcal{D}_i^{3D} \otimes \mathcal{M}_i^{3D} \rangle \quad (2)$$

where $\otimes$ denotes pixel-level multiplication. Subsequently, $\tilde{\mathcal{X}}_i$ is fed into feature extractor Ω for multimodal representation, which can be expressed as

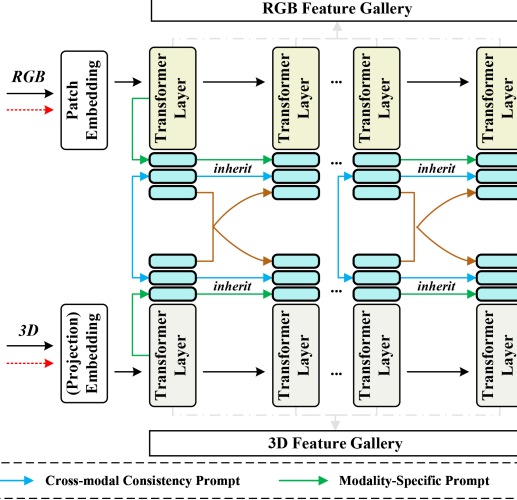
$$\langle \mathcal{F}_i^{rgb}, \mathcal{F}_i^{3D} \rangle = \Omega(\tilde{\mathcal{X}}_i). \quad (3)$$

2) *Feature-Level Missing*: Different from input-level missing, feature-level missing is designed for methods equipped with feature alignment or pre-extraction, such as memory bank and feature gallery, etc. It is worth noting that since the feature extractor Ω is frozen, feature-level missing does not result in unfair comparisons. Therefore, it is considered as pseudomodalities missing scheme, as the input dataloader remains complete, which

I. Missing Modalities Configuration



II. Cross-modal Prompt Learning



III. Symmetric Contrastive Learning

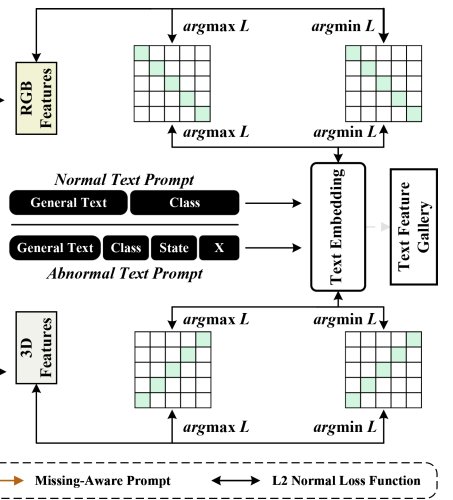


Fig. 3. Overall flowchart of our proposed framework for MISDD-MM. It consists of three serial phases: (I) MM configuration, which produced three input patterns through three modalities-missing settings; (II) CPL, which includes three specially designed prompts with colored solid lines, i.e., CCP, MSP, and MAP; (III) SCL, which performs triple-modal contrastive pretraining to generate defect detection results. Prompt injection occurs at early transformer layers, where input tokens are pretended with three prompts based on current modality availability.

can be formulated as

$$\mathcal{X}_i = \langle \mathcal{D}_i^{\text{rgb}}, \mathcal{D}_i^{\text{3D}} \rangle. \quad (4)$$

Similarly, the feature extraction of modality-complete input $\mathcal{X}_i$, defined as

$$\langle \mathcal{F}_i^{\text{rgb}}, \mathcal{F}_i^{\text{3D}} \rangle = \Omega(\mathcal{X}_i) \quad (5)$$

then missing operation is performed on the extracted feature maps $\mathcal{F}_i^{\text{rgb}}$ and $\mathcal{F}_i^{\text{3D}}$ as the following formulation:

$$\langle \widetilde{\mathcal{F}_i^{\text{rgb}}}, \widetilde{\mathcal{F}_i^{\text{3D}}} \rangle = \langle \mathcal{F}_i^{\text{rgb}} \otimes \mathcal{M}_i^{\text{rgb}}, \mathcal{F}_i^{\text{3D}} \otimes \mathcal{M}_i^{\text{3D}} \rangle. \quad (6)$$

C. Cross-Modal Prompt Learning

1) *RGB and 3D Feature Representation*: Vision transformer (ViT) [45] has strong global modeling capabilities and compatibility with text transformers, making it an ideal choice for CLIP [25] to achieve efficient cross-modal alignment. However, [24], [46] disclosed the fact that global feature representation based on Q - K self-attention actually weakens the localization ability of CLIP. Instead, Li et al. [47] proposed a consistent self-attention to build local relations for consistent semantics, which relies on the self-attention matrix VV^T . The implementation is as follows:

$$\text{Attention}(V, V, V) = \text{softmax} \left(\frac{VV^T}{\sqrt{d_k}} \right) V. \quad (7)$$

In this article, ViT based on consistent self-attention is leveraged to extract features from RGB images and depth maps, while PointMAE [48] is adopted for point cloud data. In addition, two feature galleries G^{rgb} and G^{3D} are established to store extracted features of normal training samples from RGB and 3-D modalities.

2) *Missing-Aware Cross-Modal Prompt*: Modalities-missing brings several troubles for cross-modal learning [21], [34], including: 1) Problem of establishing multimodal information consistency; 2) Learning mode transformation under different input patterns; 3) Information vacancy under modalities-missing. Correspondingly, we propose three distinct prompts to address the above issues. Note that considering that multimodal learning under MM in this article is based on dual visual modalities, we assign the same prompts to the pretrained transformer layers for RGB and 3D modalities. Below three well-designed prompts are introduced in turn.

Cross-Modal Consistency Prompt: As introduced in (1), modality-missing configurations leading three different input patterns, which makes it difficult to establish and store consistent information between RGB and 3-D modalities. To this end, a CCP is proposed to ensure a consistent buffer area exists under different inputs. Simply but effectively, CCP is a randomly initialized parameter matrix $\mathcal{P}_1 \in \mathbb{R}^{l \times d}$, where l is the predetermined length of prompt, and d represents the embedding dimensions of transformer layer. In addition, CCP is attached to the first layer in parallel feature extractors.

Modality-Specific Prompt: To cope with the difference resulting from modality-complete or modality-missing in the single modal branch, we design a MSP for RGB or 3-D modality, realizing the function of updating the prompt content based on different input patterns. Specifically, MSP is generated based on model input and is inherited as the number of layers increases, which leverages self-attention mentioned in (7) and can be formulated as

$$\mathcal{P}_2 = \Pi_{i=1}^h \text{softmax} \left(\frac{\mathcal{W}^h \mathcal{X}^m (\mathcal{W}^h \mathcal{X}^m)^T}{\sqrt{d_k}} \right) \mathcal{W}^h \mathcal{X}^m \quad (8)$$

where Π is concatenation function, while h denotes the number of heads in self-attention, $\mathcal{W}^h$ represents projection matrix, and

$\mathcal{X}^m$ is the input $\mathcal{D}^m$ of modality m in (2) or (4). Therefore, learning mode transformation under different input patterns can be adapted with by utilizing of MSP.

Missing-Aware Prompt: Information vacancy is the most serious problem caused by modalities missing, which requires each layer of the transformer to have the ability to deal with it independently [21]. Based on this observation and similar to CCP and MSP, MAP $\mathcal{P}_3$ is designed to attach to all layers. Concretely, the size and initialization of MAP are the same as CCP, but the prompts in the two branches do not share parameters. Moreover, the update of MAP in the transformer layers is cascaded, which in j th layer can be formulated as (8) by concatenating $\mathcal{P}_3$ with $\mathcal{X}^m$. The insertion of MAP compensates for the adaptability of information vacancy in cross-modal learning, avoiding feature fragmentation caused by dynamic modalities-missing.

Prompt Injection: The prompts are integrated into the modified residual attention blocks of the visual encoder, specifically at the early transformer layers (i.e., blocks where the layer index is zero and the depth of the layer falls within the predefined prompt depth, which is set to 6 by default). Specifically, at each relevant layer, a multihead attention operation is used to refine three prompts in relation to the available visual features, followed by sequence concatenation. The concatenated prompt-enhanced sequence is then passed through the standard attention and MLPs.

D. Symmetric Contrastive Learning

The overall design of SCL adopts a dual-branch structure where two visual modalities (RGB and 3D) are independently aligned with the text modality. Each branch shares the same architecture and learning mechanism, forming two parallel and symmetric pathways for vision-language contrastive learning, as illustrated in Fig. 3.

1) **Antithetical Text Prompt:** Positive sample-based prompt learning cannot achieve effective contrastive learning, so we build negative sample prompts for the text modality. Inspired by [24], [32], the semantic inversion of text can be achieved by concatenating abnormal “STATE.” As shown in Fig. 3, normal and abnormal text prompts with binary semantics are constructed in a structurally symmetric way. Specifically, the normal text prompt that fed into the text tokenizer is designed as $\mathcal{T}^n = [\text{General Text}][\text{Class}]$, where [General Text] represents sentence like “A photo of a...”, which is the same for all defect classes. On the contrary, the abnormal text prompt is implemented by affixing a “State” to a normal text prompt, such as “with crack.” In addition, to ensure its flexibility, we have respectively set fixed and learnable suffixes $[X]$ for abnormal text prompts, which are designed as $\mathcal{T}^{an} = [\text{General Text}][\text{Class}][\text{State}][X]$. Based on this design, symmetric input patterns are established for the contrastive learning objectives.

2) **Triple-Modal Contrastive Pretraining:** Based on the binary semantic embedding T^n and T^{an} generated by the antithetical text prompt, while RGB features F^{rgb} and 3-D features F^{3D} under modality missing are obtained through CPL. As can be seen from Fig. 3, due to the involvement of features of three modalities, the optimization is considered as two sets of

symmetrical vision-language contrastive learning. The contrastive loss is designed as the sum of two sub-losses, each operating between a visual modality and the binary text embeddings. Specifically, the parameters Θ of cross-modal prompts and text prompts are optimized through

$$\Theta = \arg \min [L(F^{\text{rgb}}, T^n) + L(F^{3D}, T^n) - L(F^{\text{rgb}}, T^{an}) - L(F^{3D}, T^{an})] \quad (9)$$

where L is defined as the L2 norm of F^m and T^t , where m denotes RGB or 3-D modality, and t denotes normal or abnormal text embedding.

E. Multimodal Defect Detection

All prompts are fine-tuned through CPL and triple-modal contrastive pre-training, with intermediate visual and text features stored in G^{rgb} , G^{3D} , and G^{text} . Based on these, the partial image-level score I and pixel-level map P are obtained as follows:

$$I_i^{m_1, m_2} = F_i^{m_1} \cdot (G_i^{m_2})^T \quad (10)$$

$$P_i^{m_1, m_2} = \sum_{i=1}^k (1 - F_i^{m_1} \cdot (G_i^{m_2})^T) \quad (11)$$

where m_1 and m_2 represent different modalities, and k denotes the number of layers that generated middle features. Consequently, the final detection scores S^{im} are formulated as follows:

$$S_i^{\text{im}} = \max_i \{I_i^{\text{rgb}, \text{text}}, I_i^{3D, \text{text}}\} \quad (12)$$

and the segmentation maps S^{px} are calculated as follows:

$$S_i^{\text{px}} = \max_i \{H(I_i^{\text{rgb}, \text{text}}, P_i^{\text{rgb}, \text{text}}), H(I_i^{3D, \text{text}}, P_i^{3D, \text{text}})\} \quad (13)$$

where $H(\cdot)$ denotes harmonic mean operation [22], which assign greater weight to normality prediction.

IV. EXPERIMENTS

A. Experiments Setup

1) **Datasets:** In this article, all experiments are performed on the MVtec 3D-AD [7] dataset, which includes 2656 training samples and 1197 test samples from 10 categories collected using RGB and 3-D sensors. Each sample contains a pair of RGB images and 3-D data [i.e., colored point cloud (PC)], and depth maps are generated and aligned from the corresponding PC.

2) **Implementation Details:** All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU. The proposed method is built on ViT-B/16-based [45] CLIP [25] pretrained on LAION-400 M. Multimodal configurations are listed in Table II, baseline methods including M3DM [8], Shape-Guided [9], and CFM [16] adopt RGB images and PCs as input, while AST [25], 3DSR [10], and our proposed method adopt RGB images and depth maps. Feature-level or input-level zero filling is performed on missing modality with the missing rate of η , while that of $\frac{\eta}{2}$ for each modality in missing “both” scenario. For model training, the image size is set to 240, and the model is optimized using

TABLE I
PERFORMANCE COMPARISON OF I-AUROC (%), P-AUROC (%), AND AUPRO (%) SCORES ON MVTEC 3D-AD DATASET UNDER DIFFERENT MISSING MODALITY / RATE (η)

Missing Modality	Method	Missing Rate								
		$\eta = 30\%$			$\eta = 50\%$			$\eta = 70\%$		
		I-AUROC	P-AUROC	AUPRO	I-AUROC	P-AUROC	AUPRO	I-AUROC	P-AUROC	AUPRO
RGB	M3DM ₂₀₂₃ [8]	76.72	83.65	72.15	71.02	89.87	71.59	74.52	92.32	77.12
	Shape-Guided ₂₀₂₃ [9]	77.51	88.38	69.65	71.52	87.40	61.06	78.02	88.80	64.22
	AST ₂₀₂₃ [11]	72.37	78.93	67.08	65.97	75.52	54.76	58.27	77.27	42.21
	CFM ₂₀₂₄ [16]	74.13	88.45	77.90	74.63	88.21	78.31	71.47	88.95	77.44
	3DSR ₂₀₂₄ [10]	52.46	55.80	21.30	48.78	56.89	20.91	50.90	42.50	12.39
	Ours	78.80	93.38	79.20	73.86	90.78	70.50	68.75	89.22	65.31
3D	M3DM ₂₀₂₃ [8]	70.41	78.92	29.37	68.89	73.70	47.10	68.60	77.07	65.07
	Shape-Guided ₂₀₂₃ [9]	75.31	87.77	59.35	64.84	83.51	48.29	56.09	82.65	48.27
	AST ₂₀₂₃ [11]	76.83	93.78	83.41	75.10	92.51	79.85	73.99	92.26	78.02
	CFM ₂₀₂₄ [16]	70.76	87.49	79.36	65.44	87.17	78.37	57.03	85.71	75.18
	3DSR ₂₀₂₄ [10]	50.25	55.48	25.59	54.33	55.90	27.35	47.36	58.01	27.45
	Ours	81.97	98.16	92.68	81.74	98.10	92.44	81.82	97.98	91.99
Both	M3DM ₂₀₂₃ [8]	73.03	79.00	49.42	67.72	72.89	32.33	62.39	69.77	33.99
	Shape-Guided ₂₀₂₃ [9]	75.31	91.70	73.39	62.84	88.38	61.28	54.89	85.36	53.08
	AST ₂₀₂₃ [11]	71.87	84.33	73.20	70.74	79.72	67.89	65.71	79.96	63.78
	CFM ₂₀₂₄ [16]	74.22	89.73	79.39	70.46	88.42	77.96	69.99	87.47	77.95
	3DSR ₂₀₂₄ [10]	55.57	57.00	23.26	59.22	52.25	23.15	53.15	44.28	16.78
	Ours	77.71	95.00	84.03	76.95	93.28	79.79	73.83	93.05	77.44

Missing modality = “both” represents RGB images and 3D data are missing with the rate of $\frac{\eta}{2}$ respectively.

The bold values each column indicate optimal performance.

TABLE II
DETAILED IMPLEMENTATION CONFIGURATION OF MISSING MODALITY FOR DIFFERENT MODELS

Method	2D	3D	Missing Configuration
M3DM [8]	RGB	Point Cloud	Feature-level
Shape-Guided [9]	RGB	Point Cloud	Feature-level
AST [11]	RGB	Depth Map	Input-level
CFM [16]	RGB	Point Cloud	Feature-level
3DSR [10]	RGB	Depth Map	Input-level
Ours	RGB	Depth Map	Input-level

SGD with a learning rate of 0.02, momentum of 0.9, and weight decay of 0.0005. To stabilize training and improve convergence, a cosine annealing learning rate scheduler is applied, where the learning rate gradually decreases from 0.02 to a minimum of $1e-5$ over the full number of training epochs. Except for the special descriptions mentioned above, other parameter configurations can refer to WinCLIP [22] and M3DM [8].

3) *Evaluation Metrics*: Following the commonly adopted evaluation metrics, image-level defect detection performance is evaluated with image-level area under the receiver operator curve (I-AUROC), while pixel-level defect segmentation performance is evaluated with pixel-level area under the receiver operator curve (P-AUROC) and area under the per-region overlap (AUPRO), which adopts 0.3 as the false positive rate integration threshold. Specifically, given image inputs $\mathcal{D}_i^{\text{rgb}}$, $\mathcal{D}_i^{\text{3D}}$ and the corresponding true anomaly label $\mathcal{Y}_i \in \{0, 1\}$, the prediction score S_i (regardless of image-level score S_i^{im} or pixel-level score S_i^{px}) is generated by processing multimodal inputs with uncertain missing, the AUROC score is defined as

$$\text{AUROC} = \int_0^1 \text{TPR}(f) \cdot d\text{FPR}(f) \quad (14)$$

$$\text{TPR}(f) = \frac{\sum_i \mathbb{I}(s_i \geq f \wedge y_i = 1)}{\sum_i \mathbb{I}(y_i = 1)} \quad (15)$$

$$\text{FPR}(f) = \frac{\sum_i \mathbb{I}(s_i \geq f \wedge y_i = 0)}{\sum_i \mathbb{I}(y_i = 0)} \quad (16)$$

where $\mathbb{I}$ is indicator function, and f denotes candidate threshold, which derives from the ascending sort of scores in s_i . In addition, the AUPRO score is defined as follows:

$$\text{AUPRO} = \int_0^1 \text{PRO}(f) \cdot df \quad (17)$$

$$\text{PRO}(f) = \frac{1}{N} \sum_{k=1}^N \mathbb{I}(\text{IoU}_k(f) \geq \tau), (\tau = 0.3) \quad (18)$$

$$\text{IoU}_k(f) = \frac{|R_k \cap P_k(f)|}{|R_k \cup P_k(f)|} \quad (19)$$

where N is the number of anomalous areas, R_k and P_k denote the binary map of ground truth and prediction results, and τ is the IoU threshold.

B. Main Results

1) *Quantitative Comparison*: Table I presents the experimental results of the MISDD-MM task on the MVTEC 3D-AD dataset [7], the I-AUROC, P-AUROC, and AUPRO scores under different types and rates of modalities-missing are reported for comparison. Specifically, if the missing type is set to “RGB” or “3D,” it means that the RGB image or 3-D data (point cloud or depth map) is randomly missing at a rate η , while both RGB images and 3-D data are missing with $\eta/2$ if the missing type is set to “both.” It can be seen from Table I that our proposed method has achieved state-of-the-art performance in nearly all missing settings, except for the case where 70% of RGB images are missing. Under this setting, MISDD can almost only rely on the 3-D modality, while better performing methods, such as M3DM [8], Shape Guided [9] and CFM [16], utilize color point cloud data as the 3-D modality input, which can provide a small

TABLE III
P-AUROC (%) SCORE FOR DEFECT DETECTION OF ALL CATEGORIES OF MVTEC 3D-AD DATASET WITH BOTH MODALITY MISSING

Missing Modality / Missing Rate	Methods	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
RGB / 15% 3D / 15%	M3DM ₂₀₂₃ [8]	74.52	77.77	85.02	76.02	78.49	79.36	77.78	79.36	82.79	78.84	79.00
	Shape-Guided ₂₀₂₃ [9]	88.92	91.87	95.67	85.78	92.77	86.05	92.57	95.37	95.02	93.02	91.70
	AST ₂₀₂₃ [11]	86.12	85.87	87.08	92.48	86.44	75.28	86.64	83.63	82.75	77.02	84.33
	CFM ₂₀₂₄ [16]	90.83	87.92	91.77	84.19	92.56	89.82	87.82	93.41	91.30	87.67	89.73
	3DSR ₂₀₂₄ [10]	70.04	67.96	63.45	62.36	59.92	74.08	37.44	39.52	28.14	67.07	57.00
	Ours	94.60	95.00	97.84	94.79	97.54	88.64	93.45	94.54	99.04	94.51	95.00
RGB / 25% 3D / 25%	M3DM ₂₀₂₃ [8]	73.55	72.48	79.10	69.90	71.01	71.51	68.61	74.31	76.65	71.79	72.89
	Shape-Guided ₂₀₂₃ [9]	85.10	88.63	93.26	82.08	90.73	80.87	88.83	92.62	92.60	89.09	88.38
	AST ₂₀₂₃ [11]	78.75	76.13	85.93	83.67	82.98	76.04	77.03	77.05	87.81	71.83	79.72
	CFM ₂₀₂₄ [16]	91.88	86.31	88.48	85.08	89.66	90.06	88.65	88.00	86.96	89.14	88.42
	3DSR ₂₀₂₄ [10]	29.27	56.97	65.75	39.10	48.61	42.34	39.71	67.61	70.14	62.95	52.25
	Ours	92.32	92.97	97.28	93.99	97.26	87.29	90.57	90.78	99.05	91.25	93.28
RGB / 35% 3D / 35%	M3DM ₂₀₂₃ [8]	66.36	69.40	73.88	70.62	68.16	67.48	68.30	70.10	75.62	67.79	69.77
	Shape-Guided ₂₀₂₃ [9]	76.65	86.22	91.02	79.32	89.30	76.83	86.01	90.71	91.44	86.06	85.36
	AST ₂₀₂₃ [11]	77.16	75.33	79.78	83.40	83.03	72.25	82.39	86.89	86.72	72.66	79.96
	CFM ₂₀₂₄ [16]	86.60	87.65	90.61	83.47	88.42	83.63	88.09	90.44	89.60	86.17	87.47
	3DSR ₂₀₂₄ [10]	33.75	39.86	43.21	45.58	42.29	39.10	46.92	43.60	51.62	56.82	44.28
	Ours	93.36	90.88	96.40	94.31	96.21	85.75	91.31	93.33	99.14	89.80	93.05

The bold values each column indicate optimal performance.

TABLE IV
ABLATION STUDY OF PROPOSED CPL AND SCL

Missing Modality	Method	Missing Rate							
		$\eta = 30\%$			$\eta = 50\%$			$\eta = 70\%$	
		I-AUROC	P-AUROC	AUPRO	I-AUROC	P-AUROC	AUPRO	I-AUROC	P-AUROC
RGB	Ours w/o. [CPL+SCL]	78.87	85.40	71.29	69.75	81.88	57.72	60.36	82.37
	Ours w/o. [SCL]	79.99	93.15	78.44	70.57	91.22	71.70	61.65	90.67
	Δ_1	1.12	7.75	7.15	0.82	9.34	13.98	1.29	8.30
	Ours	78.80	93.38	79.20	73.86	90.78	70.50	68.75	89.22
	Δ_2	-0.07	7.98	7.91	4.11	8.9	12.78	8.39	6.85
3D	Ours w/o. [CPL+SCL]	83.31	97.18	89.05	80.14	96.74	87.47	77.40	96.76
	Ours w/o. [SCL]	84.24	95.87	84.50	81.28	95.17	81.94	78.64	95.33
	Δ_1	0.93	-1.31	-4.55	1.14	-1.57	-5.53	1.24	-1.43
	Ours	81.97	98.16	92.68	81.74	98.10	92.44	81.82	97.98
	Δ_2	-1.34	0.98	3.63	1.6	1.36	4.97	4.42	1.22
Both	Ours w/o. [CPL+SCL]	76.74	89.19	78.79	74.47	85.56	71.51	71.77	84.17
	Ours w/o. [SCL]	78.04	93.67	80.10	74.66	92.25	76.07	72.11	92.26
	Δ_1	1.30	4.48	1.31	0.19	6.69	4.56	0.34	8.09
	Ours	77.71	95.00	84.03	76.95	93.28	79.79	73.83	93.05
	Δ_2	0.97	5.81	5.24	2.48	7.72	8.28	2.06	8.88

Δ_1 indicates the increment relative to the baseline (first row under each setting) after adopting CPL, while Δ_2 denotes that after adopting a combination of CPL and SCL.

The bold values each column indicate optimal performance.

amount of color information to compensate for the vacancy of RGB modality.

In addition, defect segmentation results of all categories with “both” modalities missing are reported in Table III. It can be seen that our proposed method achieves 95.00% , 93.28% , and 93.05% on the P-AUROC scores under different missing rates while RGB and 3-D modalities are missing simultaneously. Even with a missing rate of up to 70% , it can still achieve a P-AUROC score of over 93% , which demonstrates the effectiveness of the proposed method. Compared to other state-of-the-art MISDD methods, 3.3% , 4.86% , and 5.58% leading the runner-ups also show its progressiveness.

2) *Visualization Analysis*: Fig. 4 presents the visual comparison of defect detection results. It can be seen from Fig. 4(b) and (c), our proposed method can still effectively detect defects under high missing rates. In addition, we can also observe that methods such as M3DM and CFM have a significant dependence on RGB data, as they have inaccurate inference results when only 3-D data are available. Benefiting from CPL, cross-modal compensation enables our proposed method to adapt to different

modalities-missing situations. Due to space limitations, we only analyze the detection performance under different missing rates in the case of both modalities missing.

C. Ablation Study

We conduct ablation analysis on the two main pivots proposed in this article, i.e., CPL and SCL. The quantitative results with increments offered by CPL and SCL are reported in Table IV). Besides, visualization results of ablation studies are provided in Fig. 6. The impact analysis of the missing rate has also been reported in Fig. 5.

1) *In-Depth Analysis of CPL*: The Δ_1 value in Table IV shows the quantization increments by leveraging the CPL. Especially for defect segmentation, at least 7% P-AUROC improvement is achieved under different RGB modality-missing rates, even gaining for 21.68% AUROC score at a missing rate of 70% . However, if only a partial 3-D modality is missing, CPL fails to provide improvement, which indirectly indicates that when RGB modality is missing, 3-D modality can compensate for

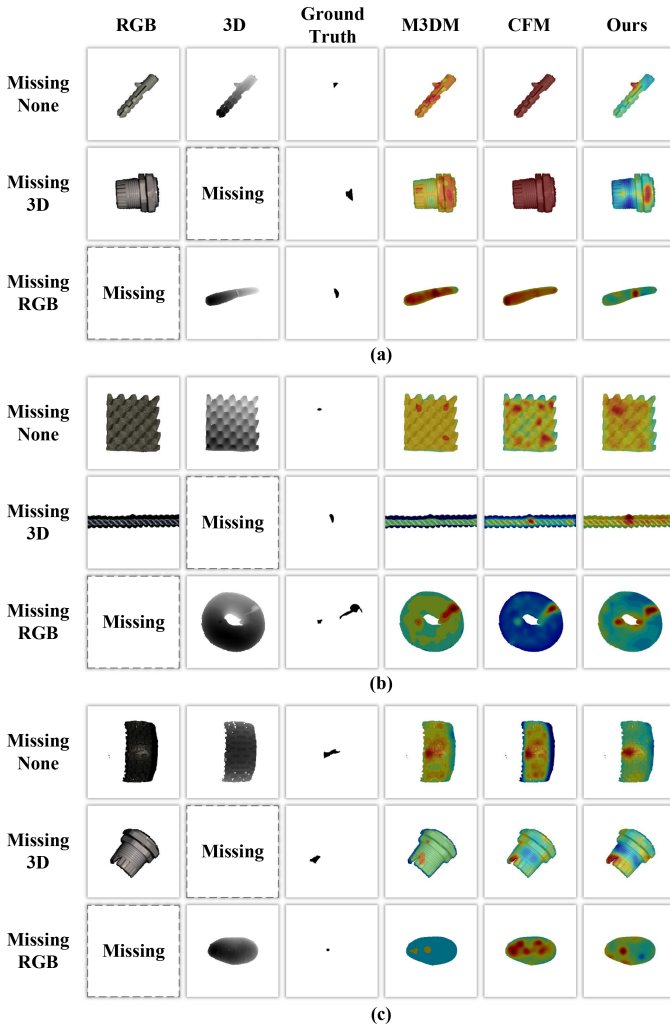


Fig. 4. Visual comparison of prediction produced by the model trained with both modalities missing under different rates. (a) Missing type = both, missing rate = 0.3. (b) Missing Type = both, missing rate = 0.5. (c) Missing type = both, missing rate = 0.7.

its absence through our proposed CPL as depth information is more vital for detecting defects with geometric variations. On the contrary, the texture information provided by the RGB modality cannot effectively improve the detection of these stereoscopic defects when the 3-D modality is missing. It also can be proved from the column 5 in Fig. 6, when the other modality is missing, the addition of CPL supplements the depth information (such as the edge of cable gland in row 2) and texture information (such as small holes on the bagel in row 3), thereby improving the confidence of defect regions.

In addition, we conducted ablation analysis of three different prompt in CPL on the MVTec 3D-AD dataset with a missing type of “both” and a missing rate of 0.3. The ablation results in Table V demonstrate that each of the three proposed prompts (CCP, MSP, and MAP) contributes positively to the model’s performance under the modality-missing condition. Notably, MSP and MAP yield larger improvements than CCP, suggesting that modality-specific and modality-aware cues play a more crucial role than simple contrastive alignment when handling

TABLE V
ABLATION OF THREE PROMPTS IN CPL ON MVTec 3D-AD DATASET

CCP	MSP	MAP	I-AUROC	P-AUROC	AUPRO
–	–	–	76.74	89.19	78.79
✓	–	–	77.23	91.46	79.85
–	✓	–	77.66	92.42	79.74
–	–	✓	77.68	92.01	79.84
✓	✓	✓	78.04	93.67	80.10

(Missing Type = “both”, Missing Rate = 0.3)

TABLE VI
PARAMETER QUANTITY STATISTICS

Entry	Setting	Params
Vision Transformer	Frozen	120.43M
Text Transformer	Frozen	91.16M
Cross-modal Consistency Prompt	Learnable	0.03M
Modality-specific Prompt	Learnable	0.2M
Missing-aware Prompt	Learnable	3.74M
Antithetical Text Prompt	Learnable	0.02M

dual-modality absence. The full configuration achieves the best results across all metrics, indicating that the proposed prompts are complementary and jointly beneficial for robust multimodal defect detection.

2) *Effectiveness of SCL*: Unlike CPL, the target task of SCL is inclined to defect recognition, as it includes CLIP-based symmetric contrastive pre-training, making it less specialized in defect localization. Table IV shows that if the RGB modality is partially missing, there are 3.29% and 7.10% improvements under missing rates of 50% and 70%, respectively. The utilization of SCL lifts the I-AUROC scores to varying degrees under other different missing settings. The visual results (last column in Fig. 6) also demonstrate the effectiveness of SCL, especially in removing false positives.

3) *Detection Performance Versus Missing Rate*: Fig. 5 shows the trend of I-AUROC scores for each category and mean value with different missing rates. Normally, when the missing type is set to both modalities missing, the I-AUROC score decreases with increasing missing rate. The modest performance drop indicates strong robustness against MM. In particular, unlike the significant decline observed when RGB data is missing, the model’s performance remains almost unchanged when only the 3-D modality is unavailable, as reflected by the near-horizontal trend line. This result discloses that the lack of 3-D modality has almost no effect on the I-AUROC score, demonstrating that most of the features used to identify defects come from the RGB modality. In contrast, the 3-D modality that mainly serves as an information supplement has little effect on identifying defects.

D. Computational Efficiency

Table VI provides a comprehensive overview of the number of parameters required by the method proposed in this article. Notably, the majority of the model parameters are contained within the frozen transformer backbone, which remains unchanged during training. The learnable components introduced by our method—including any additional layers, adapters, or

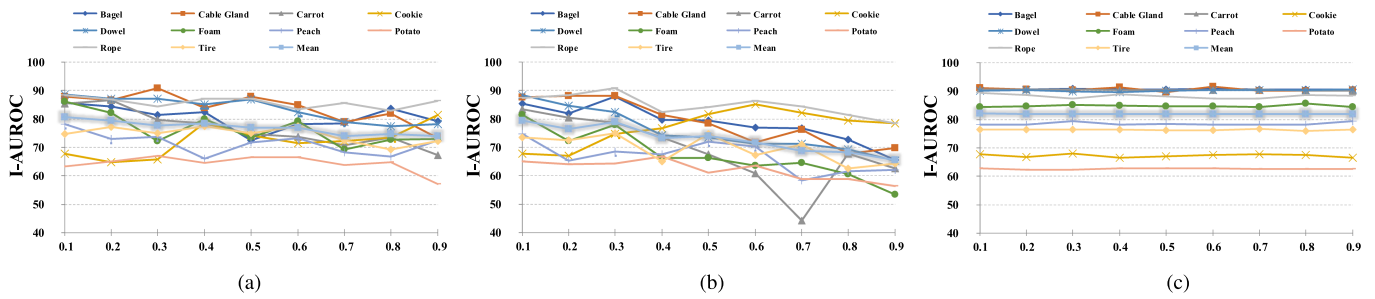


Fig. 5. Variation trend of I-AUROC scores for all categories under different missing types and rates. (a) Both modalities missing. (b) RGB modality missing. (c) 3-D modality missing.

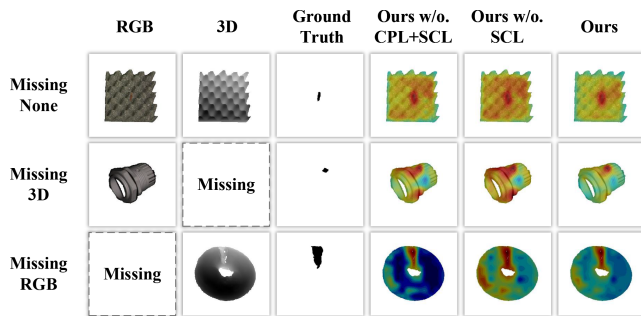


Fig. 6. Example ablation study on model trained with both modalities missing and missing rate of 70%. The last three columns show the visualization effect of the available modality (for inference) superposition prediction results.

TABLE VII
PERFORMANCE COMPARISON ON FEW-SHOT MISDD

K -shot	Method	I-AUROC	P-AUROC	AUPRO
1 shot	EasyNet [28]	0.601	0.910	0.755
	CLIP3D-AD [49]	0.733	0.969	0.893
	Ours	0.719	0.967	0.877
2 shot	EasyNet [28]	0.581	0.836	0.493
	CLIP3D-AD [49]	0.740	0.970	0.902
	Ours	0.736	0.969	0.885
4 shot	EasyNet [28]	0.586	0.860	0.565
	CLIP3D-AD [49]	0.707	0.969	0.889
	Ours	0.765	0.972	0.892

task-specific modules—constitute less than 2% of the total parameter count.

E. Few-Shot MISDD

We extend the proposed method to the few-shot MISDD task, which can be approximatively seen as a more extreme case of data missing. Specifically, we set only K samples available in the training set, and both RGB and 3D modalities missing for the others, while the test set remains complete. The experimental results are shown in Table VII, it can be seen that our proposed method has good scalability on the few-shot MISDD task, and achieves optimal performance under the 4-shot setting.

V. CONCLUSION

In this article, we propose the task termed MISDD with MM caused by uncertain sensors availability. We then offered a

framework integrated with CPL and SCL, which are leveraged to tackle the problems of learning mode transformation and information vacancy caused by dynamic modalities missing, and thus achieve text-guided fusion of dual visual modalities. Experiments under various missing settings verify the progressiveness of the proposed method.

While the proposed MISDD-MM framework demonstrates robust performance under dynamic modality-missing scenarios, several limitations remain. First, although the use of text prompts enables semantic alignment across modalities, the effectiveness of prompt generation is still constrained by the quality and diversity of manually designed templates. Second, the computational cost associated with maintaining multiple feature galleries and prompt modules, although relatively lightweight, may pose challenges for real-time deployment in resource-constrained settings. Future work will explore several directions to address these issues. One direction involves developing adaptive or generative prompt strategies to replace fixed prompt templates, thereby enhancing generalization to unseen categories or modalities.

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